

```
from sklearn.naive_bayes import BernoulliNB
```

```
X = [
    [1,1,1],
    [1,1,0],
    [1,0,1],
    [0,1,0],
    [0,1,1],
    [0,0,0]
]
Y = [
    'flu',
    'flu',
    'flu',
    'cold',
    'cold',
    'cold'
]
```

```
model = BernoulliNB()
model.fit(X,Y)
```

▼ BernoulliNB ⓘ ?

```
BernoulliNB()
```

```
patient = [[1,1,1]]
prediction = model.predict(patient)
print("predicted diagnosis: ", prediction[0])
```

```
predicted diagnosis: flu
```

```
probabilities = model.predict_proba(patient)
for diagnosis, probability in zip(model.classes_, probabilities[0]):
    print(diagnosis, "=", round(probability, 2))
```

```
cold = 0.14
flu = 0.86
```

```
patient2 = [[0,1,0]]
prediction2 = model.predict(patient2)
print("predicted diagnosis:", prediction2[0])
```

```
predicted diagnosis: cold
```

Start coding or [generate](#) with AI.